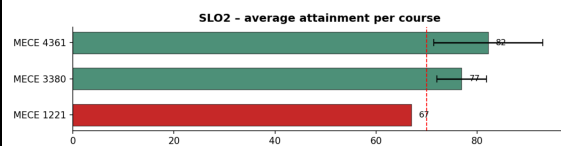
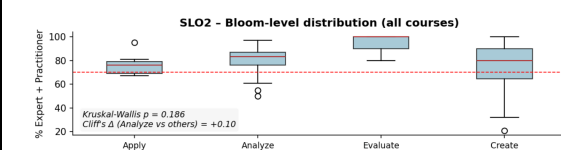
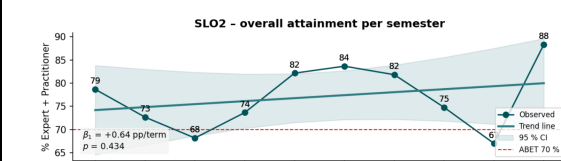


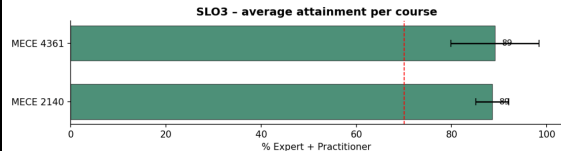
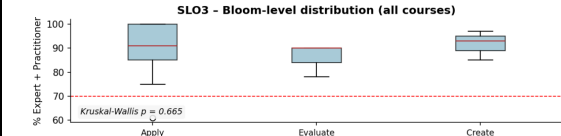
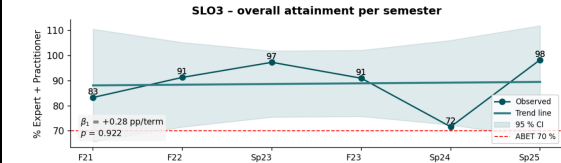
ABET Continuous Improvement

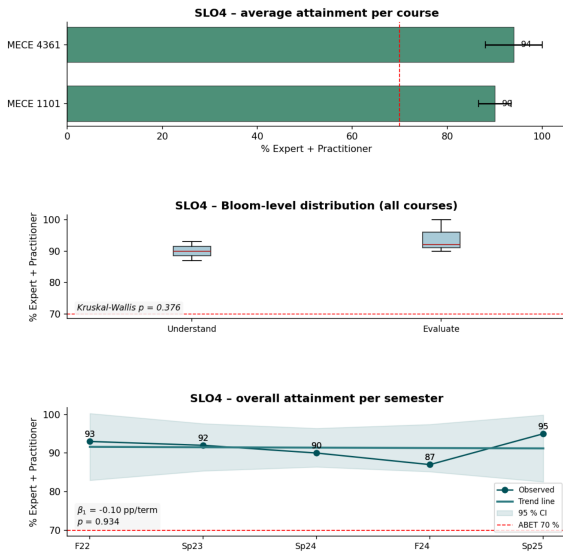
Departmental-Level Interventions (suggested)

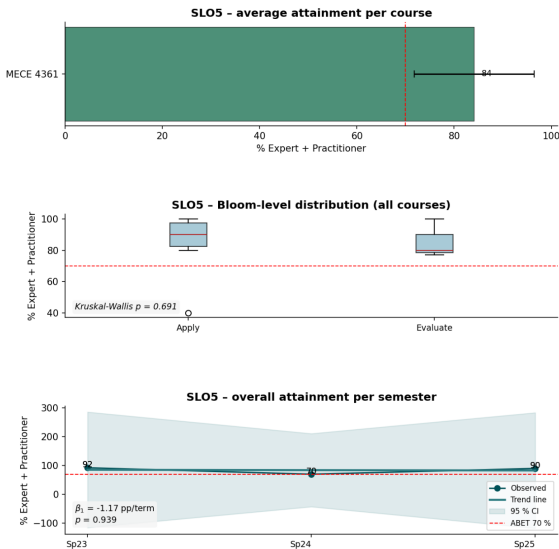
Department of Mechanical Engineering
September 9, 2025

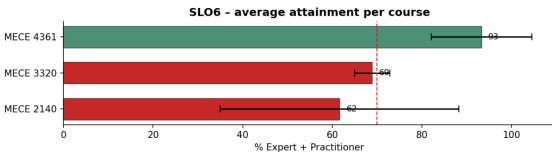
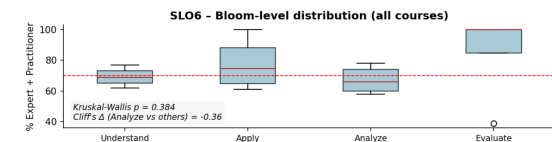
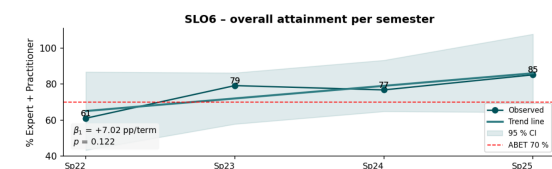
SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties																																						
SLO-1 (Problem Solving)	Students perform adequately at Understand and Apply, but attainment drops significantly at Analyze and Evaluate. Faculty observations noted that students often “plug and chug” equations without fully formulating the problem, breaking it into sub-parts, or justifying their assumptions. This weakness is confirmed by the data: multiple outliers below 50% at the Analyze/Evaluate levels, and semester-level dips below the 70% threshold in Fall 2022 and Spring 2024. Overall, the challenge lies in pushing students beyond formula recall into higher-order reasoning. SLO1 - average attainment per course <table border="1"><thead><tr><th>Course</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>MECE 3450</td><td>80</td></tr><tr><td>MECE 4350</td><td>78</td></tr><tr><td>MECE 3360</td><td>77</td></tr><tr><td>MECE 3380</td><td>74</td></tr><tr><td>MECE 3315</td><td>68</td></tr><tr><td>MECE 2340</td><td>66</td></tr><tr><td>MECE 2302</td><td>61</td></tr></tbody></table> SLO1 - Bloom-level distribution (all courses) SLO1 - overall attainment per semester <table border="1"><thead><tr><th>Semester</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>F20</td><td>81</td></tr><tr><td>Sp21</td><td>78</td></tr><tr><td>F21</td><td>78</td></tr><tr><td>Sp22</td><td>74</td></tr><tr><td>F22</td><td>70</td></tr><tr><td>Sp23</td><td>75</td></tr><tr><td>F23</td><td>79</td></tr><tr><td>Sp24</td><td>70</td></tr><tr><td>F24</td><td>76</td></tr><tr><td>Sp25</td><td>74</td></tr></tbody></table>	Course	% Expert + Practitioner	MECE 3450	80	MECE 4350	78	MECE 3360	77	MECE 3380	74	MECE 3315	68	MECE 2340	66	MECE 2302	61	Semester	% Expert + Practitioner	F20	81	Sp21	78	F21	78	Sp22	74	F22	70	Sp23	75	F23	79	Sp24	70	F24	76	Sp25	74	Problem Analysis (Apply → Analyze): Adopt a common “Given–Find–Assumptions–Approach–Units” checklist across courses; rubric checks ability to decompose problems into parts. Structured Justification (Analyze → Evaluate): Require students to explicitly justify assumptions and solution approaches in homework and exams; rubric evaluates clarity and logical reasoning. Progressive Expectations (Analyze → Evaluate): Set leveled expectations: sophomore = setup; junior = decomposition and multi-step analysis; senior = full justification and evaluation of solution.	Raise attainment at Bloom’s Analyze/Evaluate levels so ≥75% of students reach Practitioner in PI-2 and PI-3, and eliminate semester dips below the 70% threshold.	Course instructors, Curriculum Task Force, etc
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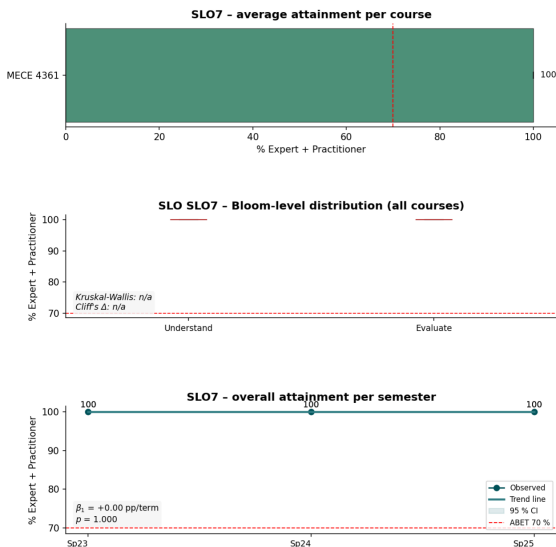
SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties																														
SLO-2 (Design)	Students performed reasonably well at the Evaluate level but showed persistent weaknesses at Apply and highly variable outcomes at Create. Faculty narratives highlighted that in Senior Design I, students struggled to articulate problem statements clearly and often had difficulty in the conceptual and embodiment design phases, especially when transitioning from broad ideas to structured design specifications. Observations noted that students frequently required more help in problem formulation and constraint analysis, and that while some produced strong projects, overall performance was uneven. The assessment data confirms these challenges: attainment dipped below the 70% threshold in Fall 2021 and Fall 2024, coinciding with the most open-ended design tasks, before recovering in Spring 2025. SLO2 - average attainment per course <table border="1"><thead><tr><th>Course</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>MECE 4361</td><td>82</td></tr><tr><td>MECE 3380</td><td>77</td></tr><tr><td>MECE 1221</td><td>67</td></tr></tbody></table> SLO2 - Bloom-level distribution (all courses)  Kruskal-Wallis $p = 0.186$ Cliff's δ (Analyze vs others) = +0.10 SLO2 - overall attainment per semester <table border="1"><thead><tr><th>Semester</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>F20</td><td>79</td></tr><tr><td>Sp21</td><td>75</td></tr><tr><td>F21</td><td>68</td></tr><tr><td>Sp22</td><td>74</td></tr><tr><td>F22</td><td>82</td></tr><tr><td>Sp23</td><td>84</td></tr><tr><td>F23</td><td>82</td></tr><tr><td>Sp24</td><td>75</td></tr><tr><td>F24</td><td>68</td></tr><tr><td>Sp25</td><td>88</td></tr></tbody></table> $\beta_1 = +0.64$ pp/term $p = 0.434$	Course	% Expert + Practitioner	MECE 4361	82	MECE 3380	77	MECE 1221	67	Semester	% Expert + Practitioner	F20	79	Sp21	75	F21	68	Sp22	74	F22	82	Sp23	84	F23	82	Sp24	75	F24	68	Sp25	88	Problem Definition Rubric (Apply → Analyze): In sophomore/junior design courses, students must submit a Problem Statement & Requirements Document. Rubric criteria: clarity of problem articulation, completeness of requirements, and identification of measurable constraints. Targets Bloom's Apply (using prior knowledge to define requirements) and Analyze (breaking down problem into constraints). Concept Development Rubric (Apply → Create): In Senior Design I, each team must generate ≥6 design concepts using at least one structured creativity method. Rubric criteria: breadth of ideas, feasibility of concepts, and quality of documentation. Targets Bloom's Apply (using structured tools) and Create (generating original design solutions). Decision-Making Rubric (Analyze → Evaluate): During Senior Design I, teams must perform a trade-off analysis using a design matrix with ≥5 evaluation criteria. Rubric criteria: appropriateness of criteria, logical evaluation process, and justification of final selection. Targets Bloom's Analyze (comparing alternatives systematically) and Evaluate (judging best solution with evidence).	≥80% of students reach Practitioner at Bloom's Evaluate/Create by senior design.	Capstone faculty
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SLO-3 (Communication & Professional Skills)	Students demonstrate strong performance in communication overall, with Apply and Create levels consistently high. However, Bloom's Evaluate is slightly weaker, and faculty narratives confirm that students often fail to critically adapt their communication to diverse audiences or fully integrate professional and technical perspectives. The dip to 72% in Sp24 highlights that performance can falter under higher expectations for professionalism and audience awareness. SLO3 - average attainment per course <table border="1"><thead><tr><th>Course</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>MECE 4361</td><td>89</td></tr><tr><td>MECE 2140</td><td>91</td></tr></tbody></table> SLO3 - Bloom-level distribution (all courses)  Kruskal-Wallis $p = 0.665$ <table border="1"><thead><tr><th>Bloom Level</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>Apply</td><td>~90</td></tr><tr><td>Evaluate</td><td>~85</td></tr><tr><td>Create</td><td>~92</td></tr></tbody></table> SLO3 - overall attainment per semester  $\beta_1 = +0.28$ pp/term $p = 0.922$ <table border="1"><thead><tr><th>Semester</th><th>% Expert + Practitioner</th></tr></thead><tbody><tr><td>F21</td><td>83</td></tr><tr><td>F22</td><td>91</td></tr><tr><td>Sp23</td><td>97</td></tr><tr><td>F23</td><td>91</td></tr><tr><td>Sp24</td><td>72</td></tr><tr><td>Sp25</td><td>98</td></tr></tbody></table>	Course	% Expert + Practitioner	MECE 4361	89	MECE 2140	91	Bloom Level	% Expert + Practitioner	Apply	~90	Evaluate	~85	Create	~92	Semester	% Expert + Practitioner	F21	83	F22	91	Sp23	97	F23	91	Sp24	72	Sp25	98	Audience Adaptation Rubric (Apply → Evaluate): Require students in mid-program courses to revise reports/presentations for different audiences (technical peer, managerial, general public); rubric assesses clarity, tone, and appropriateness. Deliverables (Analyze → Create): In junior/senior projects, require joint technical + professional deliverables (e.g., engineering report + executive summary + presentation); rubric checks for coherence. Faculty Exemplars (Evaluate → Create): Faculty will use shared rubrics and examples of excellent student communication to norm grading and provide models of effective integration.	Maintain high attainment but ensure ≥90% at Bloom's Evaluate level, with no semester falling below the 70% ABET threshold.	Project/Capstone instructors.
Course	% Expert + Practitioner																															
MECE 4361	89																															
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Bloom Level	% Expert + Practitioner																															
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SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties
SLO-4 (Ethics & Responsibility)	Students consistently score high at Bloom’s Understand, but faculty note that their ability to Evaluate ethical dilemmas is less robust. While most students can recall professional codes, many struggle when asked to weigh competing values or justify choices in ambiguous scenarios. This aligns with Bloom’s data, which shows Evaluate slightly weaker than Understand, and a semester-level dip to 87% in Fall 2024. The challenge lies in moving beyond recognition of ethics toward evidence-based evaluation of complex cases.	Case-Based Exercises (Understand → Evaluate): Introduce structured presentations/exercises in mid-program courses, with rubrics assessing ability to weigh multiple perspectives and justify decisions. Ethics-in-Context Assignments (Analyze → Evaluate): Embed discipline-specific ethical scenarios (safety, sustainability, etc) into technical courses; rubric evaluates depth of reasoning, trade-off analysis, etc. Reflective Essays (Evaluate): In capstone, require reflective essays where students defend ethical choices; rubric criteria: clarity of values, recognition of ambiguity, and justification of decisions.	Maintain overall attainment ≥90%, while ensuring ≥85% of students achieve Practitioner at Bloom’s Evaluate in ethics-related assessments.	Ethics faculty; Course instructors embedding ethics modules
				

SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties
SLO-5 (Teamwork)	Students typically succeed at Bloom's Apply level, functioning within teams to complete deliverables. However, faculty narratives and Bloom's distribution highlight that some students underperform significantly (one major outlier at Apply) and that reflection and accountability at the Evaluate level are inconsistent. Semester data confirm variability, with attainment dipping to the 70% threshold in Sp24, showing that team performance can be unstable from term to term.  SLO5 - average attainment per course MECE 4361 0 20 40 60 80 100 % Expert + Practitioner 84 SLO5 - Bloom-level distribution (all courses) 100 80 60 40 % Expert + Practitioner Kruskal-Wallis $p = 0.691$ Apply Evaluate SLO5 - overall attainment per semester 300 200 100 0 -100 % Expert + Practitioner $\beta_1 = -1.17$ pp/term $p = 0.939$ Sp23 Sp24 Sp25 Observed Trend line 95 % CI ABET 70 %	Teamwork Rubric (Apply → Evaluate): Implement a standardized rubric across project courses that evaluates contribution, collaboration, and accountability. Peer Evaluation (Evaluate): Require mid- and end-semester peer evaluations in all team-based courses; rubric criteria: fairness of assessment, evidence of contribution, and reflection on team dynamics. Reflection Component (Evaluate): Add individual reflective blurb on teamwork experiences, with rubric assessing ability to analyze group dynamics, accountability, and professional responsibilities.	Maintain ≥85% attainment in teamwork skills, with no semester falling below the 70% threshold. Ensure ≥80% of students reach Practitioner at Bloom's Evaluate level.	Faculty teaching lab/project courses; Capstone faculty.

SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties
SLO-6 (Experimentation & Analysis)	Students succeed at carrying out procedures (Apply) but struggle with Analyze (designing experiments, identifying error sources) and Evaluate (drawing conclusions, reflecting critically). Bloom’s distribution confirms weak Analyze attainment and an outlier at Evaluate. Faculty narratives highlight that lab reports often lack depth, treating labs as verification rather than inquiry. This matches course-level data, where MECE 2140 and 3320 fell below the 70% threshold, though later semesters show improvement (Sp22 = 61% → Sp25 = 85%). SLO6 - average attainment per course  SLO6 - Bloom-level distribution (all courses)  SLO6 - overall attainment per semester 	Experimental Design Rubric (Apply → Analyze): Require all mid-level labs to include experimental design components; rubric criteria: identification of variables, controls, and error sources. Lab Report Reflection (Analyze → Evaluate): Update rubrics to require explicit reflection sections where students critique data quality, compare to theory, and suggest improvements. Progressive DOE Modules (Analyze → Evaluate): Integrate short DOE (Design of Experiments) activities across sophomore–junior labs, building toward independent design in capstone labs.	Ensure ≥70% attainment at Bloom’s Analyze/Evaluate across all labs; raise average attainment in MECE 2140/3320 above threshold; maintain upward trend to ≥85% program-wide.	Lab faculty; Measurement & Instrumentation faculty;

SLO	Theme / Student Struggles	Proposed Department-Level Action	Program Assessment Target	Responsible Parties
SLO-7 (Lifelong Learning)	Current attainment is reported at 100%, but data is limited to a single course and only two Bloom's levels (Understand, Evaluate). Faculty narratives suggest that while students do complete reflective assignments, many responses remain superficial, lacking clear articulation of personal growth, future learning needs, or strategies for self-directed development. The limited dataset here makes it difficult to draw strong conclusions, but early evidence points to a need for a more structured reflection.  SLO7 - average attainment per course MECE 4361 0 20 40 60 80 100 % Expert + Practitioner 100 SLO SLO7 - Bloom-level distribution (all courses) 100 90 80 70 % Expert + Practitioner Understand Evaluate Kruskal-Wallis: n/a Cliff's Δ: n/a SLO7 - overall attainment per semester 100 90 80 70 % Expert + Practitioner Sp23 Sp24 Sp25 Observed Trend line 95 % CI ABET 70 % $\beta_1 = +0.00$ pp/term $p = 1.000$	Structured Reflection Rubric (Understand → Evaluate): Develop a common rubric for reflection assignments that scores depth of self-assessment, use of evidence, and plan for growth. Capstone Lifelong Learning Statement (Evaluate → Create): Require all capstone students to submit a “lifelong learning plan” identifying goals, resources, and strategies beyond graduation; rubric assesses specificity, realism, and creativity.	Maintain 100% attainment, but expand Bloom's coverage by ensuring data at Analyze/Create levels. Aim for ≥85% of students achieving Practitioner on structured reflection.	Capstone faculty